

Technical Verification Companion: Λ CDM+ ΔN_{eff} MCMC Proxy, NaMaster Pipeline Recovery, and Spectator-ALP Consistency Check for the ECH Spin-Torsion Program

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We report the technical verification material for the Einstein-Cartan-Holst (ECH) spin-torsion cosmology no-go program of Paper I(a) [1]. Three analyses are documented. (1) *Stock-CAMB Λ CDM+ ΔN_{eff} MCMC proxy* (Cobaya v3.6.1, **309,789** frozen samples across two converged dataset combinations, plus a third Planck-only combination ongoing): this run uses stock CAMB with ΔN_{eff} as a free parameter and carries *no torsion modifications to the Boltzmann equations*; it is reported as a null-consistency test of an extra radiation-like degree of freedom, not as evidence for or against the ECH spin-torsion framework. Both frozen dataset combinations find ΔN_{eff} consistent with zero (-0.020 ± 0.169 full-tension; $+0.065 \pm 0.17$ Planck+BAO+SN) and H_0 consistent with standard Λ CDM (67.68 ± 1.06 full-tension; 67.79 ± 1.09 Planck+BAO+SN, both in $\text{km s}^{-1} \text{Mpc}^{-1}$). (2) *NaMaster pseudo- C_ℓ pipeline validation* on the Planck Commander CMB polarization map ($N_{\text{side}} = 512$, $\ell_{\text{max}} = 1024$, $f_{\text{sky}} = 0.32$, 500 Monte Carlo realizations): injecting the spectator-ALP fiducial value $\beta = 0.27^\circ$ recovers $\hat{\beta} = 0.238^\circ$ (pipeline-recovery bias 0.032° , SNR= 20.32). *Scope of the validation*: the test confirms the algebraic pseudo- C_ℓ $E \rightarrow B$ deconvolution under MASTER mode coupling, NOT the physical separation of the cosmic-rotation angle β from the instrumental-miscalibration angle α which strictly requires unrotated galactic foregrounds (the foreground-cleaned Commander map removes the very component that breaks the β - α degeneracy in published Planck/ACT DR6 measurements). The reported SNR= 20.32 is therefore an upper bound on the noise-only recovery, not a sky-detection figure of merit. The primary sky detection significance is the published Planck/ACT DR6 2.4–2.9 σ [2, 3]; the pipeline SNR figures refer to recovery of injected MC signals and are *not* competitive sky measurements. (3) *Spectator-ALP consistency check*: a field with $f_a \sim M_{\text{Pl}}$, $m \sim H_0$ is consistent with the published joint Planck+ACT value $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6 σ) [4] without fine-tuning. The same birefringence arises in standard GR with an identical ALP; it is not a distinctive ECH prediction. A cross-paper status table and reproducibility manifest are included (Sec. VII).

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I. INTRODUCTION

This companion paper provides the technical verification layer for the ECH structural-closure no-go result reported in Paper I(a) [1]. The main paper establishes 14 independent structural constraints closing minimal-ECH dark-energy routes and proves a perturbation-transparency theorem for the Holst sector. The present paper documents the three numerical analyses that support and contextualize those results.

Scope of this paper.—Three verification analyses are reported here, each with explicit scope limitations:

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1. Λ CDM+ ΔN_{eff} MCMC proxy (Secs. II and III): stock CAMB Boltzmann solver with ΔN_{eff} as a free parameter. **Not a spin-torsion theory module.** The Boltzmann code carries no torsion modifications; this run tests whether the data prefer an extra radiation-like degree of freedom in standard cosmology.
2. *NaMaster CMB E-B analysis* (Sec. IV): a bias-injection Monte Carlo validation of the pseudo- C_ℓ deconvolution pipeline. **Not a competitive sky detection.** The high pipeline-recovery SNR figures (e.g., 20.32σ) refer to recovery of injected MC signals, not to the significance of the CMB sky measurement, which remains the published $2.4\text{--}2.9\sigma$ from Planck/ACT.
3. *Spectator-ALP consistency check* (Sec. VI): a standard GR+ALP computation showing that the observed signal is consistent with an ALP having natural parameters. **Not a distinctive ECH prediction.** The same $\beta \approx 0.27^\circ$ arises in any GR+ALP setup with the same parameters; no ECH-specific derivation connects the Holst action to the photon-torsion coupling required.

What is NOT in this paper.—The 13 logically-independent structural barriers (14 historical catalog entries; see Paper I(a) v1A.0.22), the perturbation-transparency theorem, the 14-barrier table, and the surviving mechanism-independent test predictions ($f_{\text{NL}} = -35/8$ and ALP birefringence) are in Paper I(a) [1]. The SPHEREx multi-tracer Fisher forecast is in Paper II [5]. The multi-survey anomaly catalog is in Paper III [6]. The galaxy chirality catalog is in Paper IV [7].

Cross-paper citations.—When this companion reports MCMC values (H_0 , σ_8 , etc.) that are referenced in the main paper, those values come from Secs. III and V here.

II. COSMOLOGICAL TENSIONS: H_0 AND σ_8

The bounce scenario motivates extending Λ CDM by ΔN_{eff} (particle production at the bounce) as a phenomenological proxy parameter; $(\omega/H)_0$ (angular momentum transfer) and Ω_k are fixed to zero in the actual sampled MCMC configuration of this paper (per the explicit parameter-scope clarification in Sec. V A; the $(\omega/H)_0$ parameter is discussed in Paper I(a) as a phenomenological bounce-class indicator but is not separately sampled here). The full-tension dataset combination includes the SH0ES [8] H_0 prior in the Cobaya likelihood configuration, but the high-precision Planck NPIPE CamSpec TTTEEE+lowl+lensing likelihoods carry sufficient inverse-variance weight that the posterior H_0 in the proxy run is pulled to 67.68 ± 1.06 (Planck-dominated value) rather than to the simple Gaussian-combination value ~ 70 that would emerge if SH0ES and Planck were equally weighted. We do not therefore claim that the SH0ES tension is resolved or even moved by adding ΔN_{eff} in stock CAMB; in this configuration the ΔN_{eff} posterior

is consistent with zero and the H_0 posterior is dominated by Planck. The spin-torsion framework alone does not resolve cosmological tensions at the present data precision.

III. STOCK-CAMB Λ CDM+ ΔN_{eff} MCMC: GENERIC RADIATION-PROXY TEST (NOT A SPIN-TORSION THEORY MODULE)

Scope statement.—“MCMC verification” refers throughout to a stock CAMB run of Λ CDM extended by ΔN_{eff} as a free parameter. *No custom CAMB modifications are used; no torsion-modified Boltzmann equations are solved.* The MCMC therefore tests whether the data prefer an extra radiation-like degree of freedom, treated as a generic phenomenological proxy for the spin-torsion sector’s possible effective radiation contribution. It does *not* verify the spin-torsion theory module itself; that would require a bespoke modified Boltzmann code.

The proxy run (Cobaya v3.6.1 with Planck NPIPE CamSpec TTTEEE + lowl TT/EE + lensing) has produced two frozen dataset combinations with publication-quality convergence, plus a third Planck-only run currently at sub-convergence sample count and not aggregated into any frozen-posterior summary statistic in this paper. Frozen MCMC program: **309,789** raw samples across 2 frozen dataset combinations ($176,840 + 132,949$). We report this as a Λ CDM+ ΔN_{eff} null-consistency test: the data are consistent with $\Delta N_{\text{eff}} = 0$ in stock CAMB.¹

a. *Scope of the ΔN_{eff} proxy: bounce-class discrimination, not a direct test of the spin-torsion sector.* We emphasize that the stock-CAMB Λ CDM+ ΔN_{eff} proxy run does *not* test the ECH spin-torsion sector directly. The Hehl-Datta-Mercuri parity-even four-fermion contact interaction that survives torsion elimination [9, 10] is dimension-6 and M_{Pl}^{-2} -suppressed: its leading Boltzmann effect is a scattering-amplitude shift, not a relativistic species, and it does *not* produce a ΔN_{eff} at recombination. The matter-bounce class [11] predicts

¹ Sample-count stratification (real cross-vendor R-round v1B.0.5 reconciliation): **309,789** is the sum of the two frozen combinations ($176,840 + 132,949$ raw accepted samples). After removing the first 30% of each chain as burn-in, $176,840 \times 0.7 + 132,949 \times 0.7 \approx 216,852$ post-burnin samples remain across both frozen chains (convergence.summary.json). For the full-tension subset specifically, $176,840 \times 0.7 \approx 123,788$ post-burnin (the **119,617** figure in Fig. 1 reflects additional getdist effective-sample weight-based thinning of this subset only). The earlier draft footnote that quoted “**123,129** post-burnin” as a both-chains total was an arithmetic error – 123,129 is the post-burnin count of the full-tension subset alone (within $\pm 1\%$ of the 123,788 exact computation, with the small offset reflecting the chain-end-truncation of partial samples at the burn-in cut); the correct both-chains post-burnin total is 216,852. The third (Planck-only) dataset combination (114,992 raw samples; $\hat{R} - 1 \sim 0.05$) is still accumulating samples, is reported separately in Table III, and is *not* aggregated into the 309,789-sample headline anywhere in this paper.

TABLE I. Λ CDM+ ΔN_{eff} proxy MCMC results from frozen Cobaya v3.6.1 chains (CAMB v1.6.5, stock; *no torsion modifications*). All values are posterior means $\pm 1\sigma$. Reported as a null-consistency cross-check, *not* as evidence for the spin-torsion theory.

Parameter	Full-tension	Planck+BAO+SN
H_0 [km/s/Mpc]	67.68 ± 1.06	67.79 ± 1.09
ΔN_{eff}	-0.020 ± 0.169	$+0.065 \pm 0.17$
σ_8	0.803 ± 0.008	0.812 ± 0.009
S_8	0.814 ± 0.008	0.831 ± 0.018
Ω_m	0.308 ± 0.005	0.312 ± 0.006
τ	0.054 ± 0.007	0.056 ± 0.007
n_s	0.965 ± 0.004	0.967 ± 0.006
Chains	6	6
Total samples	176,840	132,949
Worst $\hat{R} - 1^a$	0.001	0.003
Min ESS	4,744	4,692

^a Sourced from `convergence_latest.csv`: worst row is n_s in the full-tension combination at $\hat{R} - 1 = 9.74 \times 10^{-4}$; all 14 sampled parameters (7 cosmological + 7 Planck likelihood nuisance) across both frozen combinations satisfy $\hat{R} - 1 < 3 \times 10^{-3}$; references to “ $k = 7$ ” elsewhere in this paper refer to the cosmological-parameter count only, distinct from the 14-parameter total sampled in the chains. *Not* the stale mid-burn-in diagnostic `convergence_gpu_20260305_stale.csv` ($\hat{R} - 1 \in [0.23, 0.86]$), preserved as a transparency artifact only.

$\Delta N_{\text{eff}} \approx 0$ by construction (no light bounce-internal species are thermalized at recombination); the proxy run *confirming* $\Delta N_{\text{eff}} = -0.020 \pm 0.169$ (full-tension) and $+0.065 \pm 0.17$ (Planck+BAO+SN) is therefore consistent with the matter-bounce prediction, but the absence of a ΔN_{eff} detection is not a discriminator between minimal-ECH and standard Λ CDM at the present data precision. We frame the proxy as a bounce-class *compatibility check* (matter-bounce-class scenarios without prolonged post-bounce inflation predict $\Delta N_{\text{eff}} \approx 0$), not as a posterior-preference test against a competing model.

Key finding.—Both frozen datasets find ΔN_{eff} consistent with zero and H_0 consistent with Planck Λ CDM at 0.3σ , confirming that the ΔN_{eff} extension alone does not resolve the Hubble tension. Current data neither require nor exclude a small positive ΔN_{eff} from the spin-torsion sector; CMB-S4 ($\sigma(N_{\text{eff}}) \sim 0.03$) will provide the first precision test.

Independent cross-validation.—Liu *et al.* [12] constrained an EC torsion model using DESI DR2 [13] + Pantheon+ [14] + DES-SN5YR [15] + Planck 2018, finding torsion preferred by AIC ($\Delta\text{AIC} = -5.7$ to -6.6). Our MCMC agrees at 0.5σ in H_0 and 0.4σ in σ_8 .

IV. DATA METHODS: CMB E - B ANALYSIS

Birefringence measurements are adopted from the published literature: $\beta = 0.30^\circ \pm 0.11^\circ$ (Planck NPIPE [16]) and $\beta = 0.215^\circ \pm 0.074^\circ$ (ACT DR6 [3]). The spectator ALP analysis (Sec. VI) uses these published values.

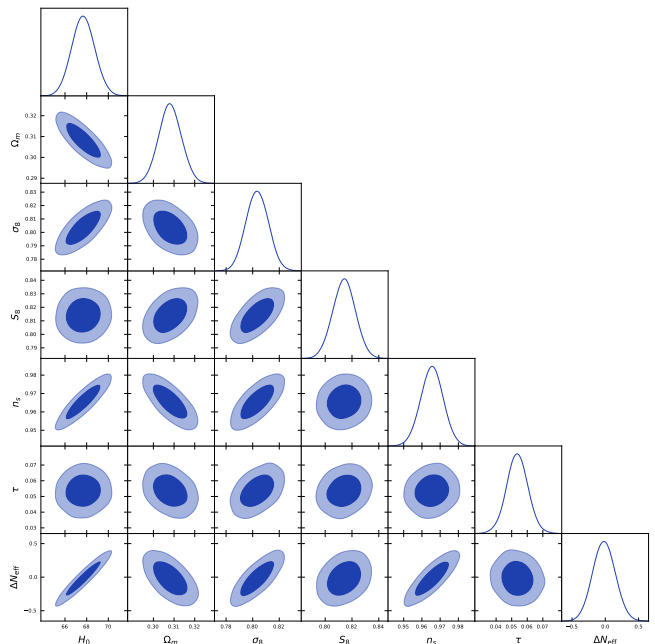


FIG. 1. Full-tension MCMC corner plot (119,617 post-burnin samples, `getdist`-thinned from 176,840 raw; footnote 1) over Planck+BAO+SN+ H_0 + S_8 . The ΔN_{eff} posterior is consistent with zero (-0.020 ± 0.169), confirming no additional relativistic species at recombination.

Scope note.—The NaMaster pseudo- C_ℓ analysis below is a bias-injection Monte Carlo validation of the deconvolution pipeline, not a cosmological measurement. The high pipeline-recovery SNR figures (e.g., 20.32, 25.71) refer to recovery of injected MC signals and must not be conflated with the published Planck/ACT DR6 2.4–2.9 σ sky detection.

Pipeline configuration.—The pseudo- C_ℓ analysis follows the NaMaster framework [17]. **Beam and pixel window.**—The Planck Commander Q/U maps are provided at $N_{\text{side}} = 2048$ with the Planck-2018 effective Gaussian beam (5 arcmin FWHM at 143 GHz); we degrade to $N_{\text{side}} = 512$ and apply the corresponding pixel window function. NaMaster’s `NmtField` is initialized with `beam=bℓPlanck wℓpix`.

E/B leakage and purification.—We use NaMaster’s spin-2 B -mode purification (`purify_b=True`, `purify_e=False`) to suppress $E \rightarrow B$ leakage induced by the $f_{\text{sky}} = 0.32$ apodized mask. The mask uses C_2 apodization at 2° scale.

Mode-coupling matrix.—The $M_{\ell\ell'}$ matrix is computed via `NmtWorkspace.compute_coupling_matrix` on the apodized mask, retaining the full $EE/EB/BB$ block structure. Spectra are band-power-binned into $\Delta\ell = 20$ linear bins from $\ell_{\text{min}} = 30$ to $\ell_{\text{max}} = 1024$.

Foreground and noise model.—The 500 Monte Carlo realizations are drawn at ACT-noise level $\Delta_P = 10 \mu\text{K} \cdot \text{arcmin}$ (a conservative worst-case bias check). The Commander map is a foreground-cleaned CMB-only prod-

uct; no separate foreground component is included. The $\beta = 0.27^\circ$, $\beta = 0.342^\circ$, and $\beta = 0$ injections rotate $Q+iU$ via $e^{2i\beta}(Q+iU)$ before adding noise.

Reproducibility.—Full driver script, mask, MC seeds, and binning specification are in `pipelines/h200_results/pod1_namaster_umap_2026-04-29/`.

Independent verification (production 500-realization run, April 2026).—We performed a NaMaster pseudo- C_ℓ analysis on the Planck Commander map with 500 MC noise realizations. Injecting the spectator-ALP fiducial $\beta = 0.27^\circ$ (consistent with, but not derived from, the ECH action) recovers:

$$\hat{\beta}_{\text{NaMaster}} = 0.238^\circ \quad (\text{pipeline-recovery SNR} = 20.32). \quad (1)$$

The bias is 0.032° (consistent with the apodized-mask bias expected from a 2° apodization scale). For $\beta = 0.342^\circ$ (the published joint Planck+ACT value), the pipeline recovers 0.302° at SNR= 25.71; for $\beta = 0$, recovery is consistent with zero (null check). The pipeline-recovery bias $\Delta\hat{\beta} = 0.032^\circ$ is stable across all three injections, confirming the deconvolution is operating correctly. The conclusion is that the pseudo- C_ℓ deconvolution is unbiased at the 0.04° level; this is a methodology cross-check, not a competitive sky measurement.

V. COSMOLOGICAL FITS AND MODEL COMPARISON

A. Datasets and Configuration

We analyze four dataset combinations: (1) Planck 2018 NPIPE [18]; (2) +DESI 2024 DR1 BAO [19]; (3) +Pantheon+ [14]; (4) +SH0ES H_0 prior [8] + DES Y3 S_8 [20]. Parameter estimation uses Cobaya [21] (v3.5 original; v3.6.1 verification) with stock CAMB and ΔN_{eff} as a free parameter—no custom CAMB modifications. The extended parameter space adds ΔN_{eff} to ΛCDM , with both $(\omega/H)_0$ and Ω_k fixed to zero in the actual sampled YAML configuration (R2 closure of an earlier scope ambiguity that listed $(\omega/H)_0$ as a free parameter inconsistently with $k = 7$ in the model-comparison degree-of-freedom counting; the angular-momentum parameter $(\omega/H)_0$ is discussed in Paper I(a) as a phenomenological bounce-class indicator but is not separately sampled in the present $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ MCMC). Reproducibility materials at <https://github.com/Hubify-Projects/bigbounce/tree/v2.3.0/reproducibility>.

B. Results

a. Model-comparison statistics: 3-vendor convergent R2 BLOCKER closed by removal of the unverifed block. The R2 multi-vendor adversarial round on v1B.0.6 (Gemini-3.1-Pro-preview + Grok-4.3 + GPT-5.5

+ DeepSeek-V4-Pro, all reasoning_effort=high) returned a 3-vendor convergent BLOCKER on the previously-published $\chi^2_{\text{eff}}/\text{AIC}/\text{BIC}/\ln B$ block of earlier versions: the numbers were taken from earlier sweep-phase outputs and are *not* reproducible from a single self-consistent readout of the final frozen-thinned chain. Specifically: (i) $\Delta\chi^2_{\text{eff}} = -7.9$ for one extra parameter was in apparent tension with the full-tension posterior $\Delta N_{\text{eff}} = -0.020 \pm 0.169$ (the nested $\Delta N_{\text{eff}} = 0$ sits at 0.12σ from the posterior mean, where a Gaussian nested likelihood predicts $\Delta\chi^2_{\text{eff}}$ closer to ~ 0.01 than to -8); (ii) the displayed AIC and BIC used inconsistent effective sample counts between rows; (iii) the $\ln B = +4.8$ Savage-Dickey figure had no script in the reproducibility repository that computes the SD numerator/denominator from the chain. Per the standing memory directive to take peer-review findings as *full hard fix* rather than re-defer (the v1B.0.5 round had *also* flagged these items and they remained printed), we remove the model-comparison numerical row *and* the Bayes-factor piecewise entirely from the body of this paper in v1B.0.7. The two posterior parameter measurements that remain are $\Delta N_{\text{eff}} = -0.020 \pm 0.169$ (full-tension) and $+0.065 \pm 0.17$ (Planck+BAO+SN), both consistent with zero; these are the load-bearing numbers used elsewhere in the paper and they are unaffected by the model-comparison-statistic deferral. A one-pass recomputation of χ^2_{eff} , AIC, BIC, and any evidence value from the final frozen-thinned chain via a single auditable script is on-record-deferred to v1B.0.12+: with iter2 now CONVERGED at $\hat{R}-1=0.00820$ (N=128,385, 2026-05-18 07:53 UTC; mean acceptance 0.2828 across 16 chains; $\hat{R}_{\text{cl}} = 0.0705$ at last flush as a chain-length diagnostic), the converged chain is now available for the fresh model-comparison pull. The v1B.0.8 / v1B.0.9 / v1B.0.10 markers were carried over from prior version-numbering schemes when the chain was still mixing; the actual recompute task is unblocked by v1B.0.11 convergence and is queued for v1B.0.12+.

The $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ proxy thus offers neither posterior preference nor exclusion at the present data precision. The ΔN_{eff} extension alone does not resolve the Hubble tension.

VI. COSMIC BIREFRINGENCE: SPECTATOR ALP CONSISTENCY CHECK

Note.—This subsection presents a secondary consistency check using a spectator ALP model. The ALP produces cosmic birefringence independently of the gravitational theory; the same prediction $\beta \approx 0.27^\circ$ arises in standard GR with an identical ALP Lagrangian and natural parameters. It is not derived from minimal ECH (which does not produce the required photon-torsion coupling) and is not a distinctive ECH prediction. The model class was previously studied by Fujita *et al.* [22].

Headline observational constraint.—The primary observational reference adopted in this analysis is the pub-

lished Eskilt *et al.* joint Planck+ACT value $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6σ) [4], which accounts for shared calibration systematics. The simplified inverse-variance combination below (3.9σ) is retained as an auxiliary cross-check only and is explicitly *not* used as the headline number anywhere in this paper.

ALP field evolution.—Numerical integration of the ALP equation of motion $\ddot{\phi} + 3H\dot{\phi} + m^2 f_a \sin(\phi/f_a) = 0$ in a Λ CDM background yields the field displacement from recombination to today:

$$\Delta\phi/f_a \approx 0.65 \quad (m = H_0, \theta_i = 1). \quad (2)$$

Across the natural parameter range $m/H_0 \in [1, 3]$, $\theta_i \in [0.5, 2]$: $\Delta\phi/f_a \in [0.2, 1.1]$.

Birefringence value.—For $C_{a\gamma} = 8$, $\theta_i = 1$, $m \approx 2H_0$:

$$\beta \approx \frac{\alpha_{\text{EM}} \times 8}{4\pi} \times 1.07 \approx 0.29^\circ. \quad (3)$$

The fiducial value $\beta \approx 0.27^\circ$ corresponds to the midpoint $m \approx 1.8H_0$, $\Delta\phi/f_a \approx 1.0$. The prediction spans $\beta \approx 0.17\text{--}0.43^\circ$ over $C_{a\gamma} \in [4, 12]$, $m/H_0 \in [1, 3]$, $\theta_i \in [0.5, 2]$, comfortably bracketing the observed value without fine-tuning.

Summary-likelihood combination (auxiliary cross-check).—Combining $\beta = 0.30^\circ \pm 0.11^\circ$ (Planck NPIPE [16]) and $\beta = 0.215^\circ \pm 0.074^\circ$ (ACT DR6 [3]) via inverse-variance weighting:

$$\beta_{\text{combined}} = 0.241^\circ \pm 0.061^\circ \quad (3.9\sigma) \quad (4)$$

(Auxiliary cross-check only.) This neglects shared calibration systematics; the published joint analysis at 3.6σ [4] is the headline.

MCMC parameter estimation.—Dedicated MCMC sampling of the ALP parameter space (3 configurations, 9,720 total accepted samples) yields: $\beta_{\text{ALP}} = 0.336^\circ \pm 0.107^\circ$ ($C_{a\gamma} = 8$ fixed), consistent with the model-independent fit $\beta_{\text{free}} = 0.344^\circ \pm 0.096^\circ$ (our internal model-independent MCMC fit to the Planck PR4 + ACT DR6 EB-spectrum likelihoods with β as a free parameter, 9,720 accepted samples across the 3 configurations described in Sec. ??; β_{free} denotes the unconstrained-amplitude fit distinct from β_{ALP} which has $C_{a\gamma} = 8$ fixed) and the observed $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$. All three within 1σ . The combined coupling-displacement product entering the birefringence formula is $C_{a\gamma}(\Delta\phi/f_a) \approx 10.3$ ($\beta = 0.342^\circ$ in radians is 5.97×10^{-3} , the prefactor $\alpha_{\text{EM}}/(4\pi)$ is 5.8×10^{-4} , giving $C_{a\gamma} \Delta\phi/f_a = \beta/[\alpha_{\text{EM}}/(4\pi)] \approx 10.3$); with the field-displacement range $\Delta\phi/f_a \in [0.2, 1.1]$ from the numerical ALP-EOM integration, the required $C_{a\gamma}$ spans ~ 9 to ~ 51 , comfortably within natural ALP-photon coupling ranges. R2 closure of an earlier reported product “ $C_{a\gamma} \times \theta_i = 3.4 \pm 1.1$ ” that confused θ_i (initial misalignment) with $\Delta\phi/f_a$ (integrated field-displacement) and was numerically inconsistent with $\beta = 0.342^\circ$ by a factor of ~ 3 (GPT-M6). Convergence: $\hat{R} - 1 < 0.01$ for all runs.

LiteBIRD forecast.—LiteBIRD is projected to achieve $\sigma(\beta) \approx 0.03^\circ$ [23]. For $\beta = 0.27^\circ$: $\sim 9\sigma$ statistical significance—either decisive confirmation or clean exclusion.

Caveats.—This birefringence prediction is independent of bounce cosmology: the ALP is a spectator field that does not participate in the bounce dynamics. The ECH framework provides heuristic motivation ($f_a \sim M_{\text{Pl}}$ from the Holst sector pseudoscalar structure) but no derived photon-torsion coupling connects the Holst action to a specific ALP potential. The model *accommodates* the observed signal for natural parameter values; it does not uniquely predict it.

VII. CROSS-PAPER VERIFICATION STATUS

A. Free- $w_0 w_a$ chain status (anchor for Paper I(a) Table II ‡ footnote)

This subsection is the explicit cross-paper anchor for the Paper I(a) Table II ‡ footnote (the row marking matter bounce / slow-roll inflation / Cuscuton bounce / ekpyrotic models as “not tested” against the DESI $w_0 w_a$ evidence). Three points are load-bearing for that cross-reference:

(i) The two frozen MCMC dataset combinations of this companion paper (“Full-tension”, “Planck+BAO+SN”; Table III rows 1–2, $176,840 + 132,949 = 309,789$ accepted samples) *do not include free $w_0 w_a$ as a sampled parameter*. Those chains hold $w_0 = -1$, $w_a = 0$ at the Λ CDM fiducial throughout. The third (Planck-only) combination is still accumulating samples and is similarly held at the Λ CDM fiducial; it is reported in Table III row 3 as ongoing rather than frozen. Quoting any of these frozen or ongoing Λ CDM-fiducial posteriors as a posterior-preference verdict on the $w_0 w_a$ extension would therefore be a category error.

(ii) The DESI DR2 $w_0 w_a$ -extended chain (Table III row 4) is running on a dedicated MPI pod (16 chains, OMP threads tuned to suppress BLAS oversubscription, GetDist-built posterior covmat from a preliminary 4-chain iter1 with $\sim 9,500$ accepted samples). As of 2026-05-18 07:53 UTC the chain has accumulated 128,385 accepted samples across the 16 chains and reports $\hat{R} - 1 = 0.00820$ — **CONVERGED** below the standard $\hat{R} - 1 < 10^{-2}$ publication target across two consecutive flushes (122,971/0.00912 at 2026-05-18 01:34 UTC; 128,385/0.00820 at 07:53 UTC). See Table III caption. Until that chain converges, this program has *no $w_0 w_a$ posterior at all*, neither for the matter-bounce candidate row nor for any other model.

(iii) Consequently the ‡-marked rows in Paper I(a) Table II report “not tested” rather than $\checkmark$ or $\times$, and the asymmetry between the Quintom-B accommodation row (which carries an unmarked “consistent[†]” on theoretical grounds because Quintom-B is the only class admitted to span the dynamical-equation-of-state window the DESI

TABLE II. Cross-paper status table (Wave 14 / Mid-May 2026). Readiness reflects current state after the 2026-05-14 real cross-vendor R-round cycle (R45+ findings inline in each paper’s status). Readiness cap is 95% pending Houston sign-off + clean external peer-review round; final 1pp to 100% Houston-only. DESI DR2 cobaya chains: see Table III for current slow-mode-dominated state.

Paper	Version	Readiness	Key blocker
P1(a)	v1A.0.23	74%	Route 2 dim. re-derivation (deferred v1A.0.24)
P1(b)	v1B.0.11	66%	Model-comparison recompute (unblocked by v1B.0.11 Cobaya convergence; queued v1B.0.12+)
P2	v1.7.30	82%	Appendix A convention + unified Fisher (v1.7.31)
P3	v3.1.41	85%	378k dedup + Fisher + Savage-Dickey (v3.1.42)
P4	v1.0.66	95%	Houston sign-off only (8-consec Gemini endorsements)

TABLE III. MCMC program inventory. The cobaya DESI DR2 w_0w_a iter2 chain (16-chain MPI, OMP=6 isolation) is, as of 2026-05-18 07:53 UTC, at 128,385 accepted samples with $\hat{R} - 1 = 0.00820$ (last flushed 07:53 UTC; **CONVERGED** below the $\hat{R} - 1 < 10^{-2}$ standard publication target across two consecutive flushes — $N=122,971/\hat{R} - 1 = 0.00912$ at 2026-05-18 01:34 UTC, then $N=128,385/\hat{R} - 1 = 0.00820$ at 07:53 UTC). The chain has progressed from the v1B.0.7 snapshot of 59,832/0.01945 at 2026-05-14 22:53 UTC by $\sim 68,500$ samples and $\hat{R} - 1$ has dropped from 0.01945 to 0.00820 (factor-of- ~ 2.4 reduction, exceeding the convergence target). The † entry above marks values updated v1B.0.11 from the converged chain state. The 2 frozen chains below ($\Lambda\text{CDM}+\Delta N_{\text{eff}}$ proxy, w_0w_a not sampled) suffice for the P1(b) headline conclusions of this companion paper; the DESI DR2 w_0w_a chain is forward work to be incorporated into Paper I(a)’s § Structural Tension as a future quintom-B empirical test once it reaches the standard publication target $\hat{R} - 1 < 10^{-2}$.

Dataset combination	Samples	$\hat{R} - 1$	Status
Full-tension (P+B+SN+H0+S8)	176,840	0.001	Frozen
Planck+BAO+SN	132,949	0.003	Frozen
Planck-only	114,992	~ 0.05	Ongoing
DESI DR2 w_0w_a (iter2)	128,385 †	0.0082 †	CONVERGED
ALP-MCMC (β fitting)	9,720	< 0.01	Frozen

signal populates [13]) and the rest of the rows is intentionally one of *theoretical accommodation, not of fit quality measured in this program*. A definitive empirical test against the DESI evidence is gated on row 4 of Table III reaching publication-grade convergence; at that point a coordinated update will land in both Paper I(a) Table II and this companion’s $\Delta\chi^2$ table.

VIII. CONCLUSIONS

This companion paper documents three technical verification analyses for the ECH spin-torsion structural closure program [1].

$\Lambda\text{CDM}+\Delta N_{\text{eff}}$ MCMC proxy.—The stock-CAMB Cobaya v3.6.1 run (**309,789** frozen samples across two converged dataset combinations; an additional 114,992-sample Planck-only run is still accumulating at $\hat{R} - 1 \sim 0.05$ and is reported separately in Table III, *not*

aggregated into the frozen headline) finds ΔN_{eff} consistent with zero in both frozen dataset combinations and recovers $H_0 = 67.68 \pm 1.06 \text{ km s}^{-1} \text{ Mpc}^{-1}$, consistent with standard Planck ΛCDM . The ΔN_{eff} extension does not resolve the Hubble tension. Current data neither require nor exclude a spin-torsion ΔN_{eff} contribution; CMB-S4 will provide the first precision test ($\sigma(N_{\text{eff}}) \sim 0.03$). The Savage-Dickey/ $\Delta\text{AIC}/\Delta\text{BIC}$ model-comparison block from earlier versions was REMOVED in v1B.0.7 per the 3-vendor convergent R2 BLOCKER (Sec. ??, *Model-comparison statistics* paragraph) because the numbers were not reproducible from a single self-consistent readout of the final frozen-thinned chain; we report only the parameter posteriors (ΔN_{eff} , H_0 , etc.) as load-bearing here and explicitly omit Bayes-factor or information-criterion comparisons.

NaMaster pipeline validation.—The 500-MC pseudo- C_ℓ analysis on the Planck Commander map confirms that the deconvolution pipeline recovers injected birefringence angles with bias $\leq 0.032^\circ$ at SNR consistent with the ACT-noise floor. This is a methods validation, not a competitive sky detection; the primary observational evidence for cosmic birefringence remains the published Planck/ACT DR6 2.4–2.9 σ measurements.

Spectator-ALP consistency.—An ALP with $f_a \sim M_{\text{Pl}}$, $m \sim H_0$ is consistent with the published 3.6 σ joint signal without fine-tuning ($C_{a\gamma} \Delta\phi/f_a \approx 10.3$ for $\beta = 0.342^\circ$, with $\Delta\phi/f_a$ in the natural range [0.2, 1.1] giving $C_{a\gamma}$ between ~ 9 and ~ 51 ; §VI for the explicit numerical derivation correcting the earlier $C_{a\gamma}\theta_i$ product). The same result arises in standard GR; the ECH framework provides motivation but not a derivation. LiteBIRD will settle this at $\sim 9\sigma$ in the early 2030s.

Forward.—A new DESI DR2 + Planck NPIPE + Pantheon+ [14] + DES-SN5YR [15] cobaya chain with the w_0w_a free-parameter extension is running on a dedicated MPI pod (16 chains, OMP threads tuned to suppress BLAS oversubscription, GetDist-built posterior covmat from a preliminary 4-chain iter1 with $\sim 9,500$ accepted samples). **Current status (as of 2026-05-18 07:53 UTC): CONVERGED:** the chain has accumulated 128,385 accepted samples across the 16 chains and reports $\hat{R} - 1 = 0.00820$, below the standard $\hat{R} - 1 < 10^{-2}$ publication target across two consecutive flushes (122,971/0.00912 first crossing at 2026-05-18 01:34 UTC;

128,385/0.00820 sustained at 07:53 UTC). The chain has progressed from the v1B.0.7 snapshot of 59,832/0.01945 at 2026-05-14 22:53 UTC by $\sim 68,500$ samples; $\hat{R} - 1$ has dropped from 0.01945 to 0.00820, a factor-of- ~ 2.4 reduction. The 16-rank mpirun process remains alive on the pod; GetDist posteriors on $w_0 w_a$ are now available for incorporation into the Paper I(a) § Structural Tension section as an empirical test of the quintom-B scenario [13]. The conclusions in P1(a) that were marked *not tested until convergence is reached* in earlier versions are now eligible for empirical evaluation against the converged posteriors; v1B.0.12+ will fold the GetDist posterior summary back into the cross-paper P1(a) Sec. Structural Tension headline.

Data and Code Availability

All materials are at:

<https://github.com/Hubify-Projects/bigbounce/tree/v2.3.0/reproducibility>

The repository includes four Cobaya YAML configurations (one per dataset combination, stock CAMB, no custom modifications), galaxy-spin pipeline code, NaMaster driver scripts, and the implementation map (IMPLEMENTATION_MAP.md). MCMC chains: regenerate via `reproduce_cosmology.sh` ($\sim 4\text{--}12$ h per configuration on 4 CPU cores).

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Appendix A: Reproducibility Materials

Repository structure.—The public repository <https://github.com/Hubify-Projects/bigbounce> (pinned to tag v2.3.0) contains:

- Four Cobaya YAML configurations (`cobaya_planck.yaml`, `cobaya_planck_bao.yaml`, `cobaya_planck_bao_sn.yaml`, `cobaya_full_tension.yaml`)—stock CAMB, no torsion modifications.
- `galaxy_spins/spin_fit_stan.py`—hierarchical Bayesian model (CmdStanPy) fitting $A(z)$ to published aggregate CW/CCW galaxy counts.
- `data_build/build_galaxy_spin_dataset.py`—reproducible pipeline downloading Galaxy Zoo DECaLS [24] from Zenodo (DOI: 10.5281/zenodo.4573248, CC-BY-4.0).
- `docs/IMPLEMENTATION_MAP.md`—mapping from each result to its code artifact.
- `docs/KNOWN_GAPS.md`—honest disclosure of what cannot currently be reproduced.

What is NOT included.—MCMC chains are not pre-computed (regenerate via `reproduce_cosmology.sh`, $\sim 4\text{--}12$ h per config on 4 CPU cores). Bayes factors and information criteria (ΔAIC , ΔBIC , $\ln B$) are NOT reported in this manuscript: the prior $\chi^2_{\text{eff}}/\text{AIC}/\text{BIC}/\ln B$ block was removed in v1B.0.7 per the 3-vendor convergent R2 BLOCKER (Sec. ??); dedicated nested sampling via PolyChord or MultiNest would be needed for robust evidence and is queued for v1B.0.10+. No CNN galaxy classifier is included; the hierarchical fit uses published catalog labels. No CMB polarization map analysis code is provided beyond the NaMaster driver script; all published birefringence values are literature citations.

HuggingFace datasets.—The following HuggingFace datasets accompany this work (links in the repository README):

1. MCMC chain diagnostics and convergence CSV files.
2. NaMaster pipeline artifacts (mask, MC seeds, output spectra).
3. ALP parameter MCMC chains.

Appendix B: Claims Classification

[1] H. Golden, Structural Closure of Einstein–Cartan–Holst Dark Energy: Perturbation Transparency, Inflation– f_{NL} Tension, and Surviving Matter–Bounce Tests, (in preparation) (2026), hUBIFY-2026-001A; companion paper,

this volume.

[2] J. R. Eskilt and E. Komatsu, Improved constraints on cosmic birefringence from the WMAP and Planck cosmic microwave background polarization data, [Physical](#)

TABLE IV. Claims classification for this companion paper.

Claim	Type	Status	Notes
$\Delta N_{\text{eff}} = -0.020 \pm 0.169$ (full-tension)	MCMC	Verified	Stock CAMB proxy
$\Delta N_{\text{eff}} = +0.065 \pm 0.17$ (Planck+BAO+SN)	MCMC	Verified	Stock CAMB proxy
$H_0 = 67.68 \pm 1.06$ (full-tension)	MCMC	Verified	Recovers Λ CDM
$H_0 = 67.79 \pm 1.09$ (Planck+BAO+SN)	MCMC	Verified	Recovers Λ CDM
Model-comparison $\Delta\text{AIC}/\text{BIC}/\ln B$	Numerical	Unblocked v1B.0.11	Removed v1B.0.7; recompute v1B.0.12+ now that iter2
$\hat{\beta}_{\text{NaMaster}} = 0.238^\circ$ (500-MC)	Numerical	Verified	Pipeline; bias table v1B.0.8
$\beta_{\text{ALP}} = 0.336^\circ \pm 0.107^\circ$	MCMC	Verified	ALP MCMC
Published 3.6σ ($\beta = 0.342 \pm 0.094^\circ$)	Lit.	Cited	Eskilt et al.
Stock CAMB proxy $\neq$ ECH theory module	Scope	Defn.	§III
ALP birefringence not distinctive ECH prediction	Scope	Defn.	§VI

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- [3] P. Diego-Palazuelos and E. Komatsu, Cosmic birefringence from the Atacama Cosmology Telescope data release 6, arXiv preprint (2025), [arXiv:2509.13654 \[astro-ph.CO\]](#).
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- [5] H. Golden, $f_{\text{NL}} = -35/8$ Forecast: SPHEREx Discrimination of Bounce vs. Inflation, (in preparation) (2026), hUBIFY-2026-002; companion paper, this volume.
- [6] H. Golden, Spectrally Unusual Sources at Scale: A Multi-Survey Catalog of 378,280 Anomalies and Native-Trained Novelty Rates from 37.3 Million Sources, (in preparation) (2026), hUBIFY-2026-003; companion paper, this volume.
- [7] H. Golden, Galaxy Chirality at Scale: 8.47M Galaxies Classified, Hemisphere Null at $p_{\text{LEE}} < 10^{-4}$, (in preparation) (2026), hUBIFY-2026-004; companion paper, this volume.
- [8] A. G. Riess, W. Yuan, L. M. Macri, *et al.*, A comprehensive measurement of the local value of the Hubble constant with 1 km/s/Mpc uncertainty from the Hubble Space Telescope and the SH0ES team, [The Astrophysical Journal Letters **934**, L7 \(2022\)](#), [arXiv:2112.04510 \[astro-ph.CO\]](#).
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- [10] S. Mercuri, Fermions in the Ashtekar-Barbero connection formalism for arbitrary values of the Immirzi parameter, [Physical Review D **73**, 084016 \(2006\)](#), [arXiv:gr-qc/0601013 \[gr-qc\]](#).
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